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Stacking-induced type-II quantum spin Hall insulators with high spin Chern number in unconventional magnetism

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ABSTRACT

While stacking two type-I quantum spin Hall insulators typically results in a trivial insulator, the behavior of the assembly of type-II quantum spin Hall insulators remains unexplored. In this article, based on calculations of a lattice model, we demonstrate that stacking two type-II quantum spin Hall insulators does not yield a trivial insulator but instead forms a nontrivial quantum spin Hall insulator with a high spin Chern number. In this phase, two pairs of topological edge states with opposite chirality and polarization coexist at the boundary. Our calculations further reveal that the quantized spin Hall conductivity of the bilayer is twice that of the monolayer. When $U(1)$ symmetry is present, the high spin Chern number phase remains stable; when $U(1)$ symmetry is broken, it persists over a broad parameter range. Furthermore, based on first-principles electronic structure calculations, we demonstrate that bilayer Nb_2SeTeO is a type-II quantum spin Hall insulator with a high spin Chern number in altermagnetism and unconventional compensated magnetism. Moreover, extending this strategy to multilayer stacks naturally leads to a quantum spin Hall insulator with a higher spin Chern number. Our work not only deepens the distinction between type-I and type-II quantum spin Hall insulators, but also offers a route toward realizing highly quantized spin Hall conductivity.

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1. Introduction

The proposal of the conventional (type-I) quantum spin Hall insulator (QSHI) has sparked widespread interest in topological phases and materials [1–10]. A type-I QSHI is known to have a bulk band gap and topological Dirac edge states protected by time-reversal symmetry at the boundaries (Fig. 1a). Its topological invariant is characterized by the Z_2 index. Stacking two type-I QSHIs typically yields a trivial insulator with $Z_2 = 0$, as illustrated in Fig. 1b.

Recently, type-II QSHIs have been theoretically proposed [11,12]. Unlike type-I QSHIs, type-II QSHIs manifest distinct spin-dependent edge states: the spin-up topological chiral edge state connects the spin-up bulk bands, while the spin-down topological chiral edge state connects the spin-down bulk bands (Fig. 1c).

Because the opposite spin-polarized bands reside in distinct regions of the Brillouin zone, type-II QSHIs do not require time-reversal symmetry for protection; instead, their topological invariant is characterized by the spin Chern number [4]. Given that type-II QSHIs necessitate anisotropic spin splitting, they can be realized in altermagnetic and unconventional compensated magnetic materials [11]. A natural question then arises: does stacking two type-II QSHIs also yield a trivial insulator, like the case of type-I QSHIs? Since type-II QSHIs are classified by the spin Chern number, stacking two QSHIs with a spin Chern number of 1 may result in a type-II QSHI with a spin Chern number of 2. Accordingly, the boundary of such a system should host two pairs of edge states with opposite chirality and spin polarization (Fig. 1d). An important open question remains whether such a type-II QSHI with a high spin Chern number, obtained by stacking two type-II QSHIs with spin Chern number 1, can indeed be realized.

In this article, we demonstrate—using an altermagnetic lattice model—that stacking two type-II QSHIs yields a type-II QSHI with a high spin Chern number. Furthermore, based on first-principles electronic structure calculations, we predict that bilayer Nb_2SeTeO

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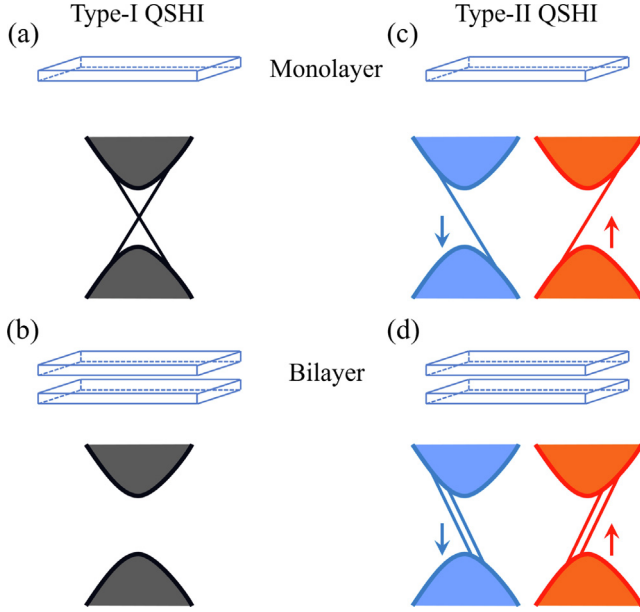


Fig. 1. Schematic of the edge states in QSHIs. (a) Monolayer and (b) bilayer type-I QSHIs. (c) Monolayer and (d) bilayer type-II QSHIs.

is a type-II QSHI with a high spin Chern number in both the altermagnetic phase and the unconventional compensated magnetic phase. The boundary of the bilayer type-II QSHI with a high spin Chern number hosts two pairs of edge states with opposite chirality and spin polarization, and its quantized spin Hall conductivity (SHC) is exactly twice that of a monolayer type-II QSHI. Moreover, the bilayer type-II QSHI phase with a high spin Chern number remains stable over a wide parameter range, irrespective of whether or not $U(1)$ symmetry is present.

2. Lattice model

To study the properties of stacking two type-II QSHIs in a bilayer lattice model, we first need to realize the type-II QSHI phase in a monolayer lattice model, which we have already constructed in our previous work [11,13,14]. For clarity, we briefly review the monolayer model here. It is a square lattice with two sublattices (one layer in Fig. 2a). The corresponding Hamiltonian reads

$$H_l = \sum_{d_i,j} \left[t^d C_{1,j}^\dagger C_{2,j+d_i} + C_{1,j}^\dagger (i\lambda^d \cdot \sigma) C_{2,j+d_i} + \text{h.c.} \right] + \sum_{\alpha,j} \left[t_\alpha^x C_{\alpha,j}^\dagger C_{\alpha,j+\mathbf{x}} + t_\alpha^y C_{\alpha,j}^\dagger C_{\alpha,j+\mathbf{y}} + \text{h.c.} \right] + \sum_{\alpha,j} \mathbf{m}_\alpha \cdot \sigma C_{\alpha,j}^\dagger C_{\alpha,j}, \quad (1)$$

where $C_{\alpha,j}^\dagger = (C_{\alpha,j1}^\dagger, C_{\alpha,j2}^\dagger)$ is the electron creation operator, t^d and $t_\alpha^{x,y}$ represent the nearest-neighbor hopping along $\mathbf{d}_i$ and the next-nearest-neighbor hopping along $\mathbf{x}, \mathbf{y}$, respectively. As illustrated in Fig. 2a, these vectors are defined as $\mathbf{d}_{1,4} = \pm \frac{1}{2}(\mathbf{x} + \mathbf{y})$ and $\mathbf{d}_{2,3} = \pm \frac{1}{2}(\mathbf{x} - \mathbf{y})$, where $\mathbf{x} = a_1 \hat{x}, \mathbf{y} = a_2 \hat{y}$ denote primitive lattice vectors with constants a_1, a_2 along the unit vectors $\hat{x}, \hat{y}$, respectively. For simplicity, we set $\lambda^{d_1} = \lambda^{d_3}, \lambda^{d_2} = \lambda^{d_4}$, and $\lambda^{d_1} = -\lambda^{d_2} = \lambda$, which represent the spin-orbit coupling (SOC) parameters. The $\mathbf{m}_\alpha = (-1)^z \mathbf{m}$ stands for the static collinear AFM order, σ_0 and σ are the identity matrix and Pauli matrices, respectively.

In this monolayer lattice model, two sites with opposite magnetic moments are located at the space-inversion invariant positions, which endows the model with spin symmetry $\{C_2^\perp | I\}$ while

breaking the spin symmetry $\{C_2^\perp | I\}$. Moreover, the next-nearest-neighbor hopping satisfies $t_1^x \neq t_2^x$ and $t_1^y \neq t_2^y$ but $t_1^x = t_2^x$ and $t_1^y = t_2^y$, which breaks spin symmetry $\{C_2^\perp | \tau\}$ while preserving spin symmetry $\{C_2^\perp | C_{4z}\}$ and $\{C_2^\perp | M_{xy}\}$. Consequently, these conditions give rise to d -wave altermagnetism in the monolayer lattice model. In addition, investigating the topological properties requires a band inversion in this monolayer lattice model, which can be obtained by imposing the condition $2|t_1 - t_2| > |\mathbf{m}|$ with $t_\alpha^x = t_\alpha$ and $t_1 = -t_2$. In the absence of SOC, the model can realize a type-I bipolarized spinful Dirac semimetal phase [14]. Upon including SOC, the gap opened in the bipolarized spinful Dirac semimetal drives the model into a type-II QSHI [11].

We first consider the $\mathbf{m} \parallel \lambda$ ($\mathbf{m} = m_z \hat{z}$) case. The corresponding monolayer lattice model has $U(1)$ symmetry, leading to spin S_z be a good quantum number. With SOC, the bipolarized spinful Dirac semimetal transitions into an insulator, as shown in Fig. 2b. As is well known, a massive spinful Dirac point in two dimensions contributes π Berry phase. In this monolayer lattice model, the C_{2z} symmetry ensures that the two spinful Dirac points along the X-M or Y-M boundary contribute the same Berry phase. Specifically, the pair of spinful Dirac points with spin-down polarization on the X-M boundary contributes -2π Berry phase, corresponding to Chern number $C_1 = -1$, while the pair of spinful Dirac points with spin-up polarization on the Y-M boundary contributes 2π Berry phase, giving $C_1 = 1$. Thus, the spin Chern number is $C_s = \frac{1}{2}(C_1 - C_2) = 1$. The corresponding SHC is quantized as $\sigma_{xy}^z = C_s e/(2\pi) = e/(2\pi)$ (computational details are provided in the Supplemental material), which is consistent with our calculation results (Fig. 2b). Moreover, open-boundary-condition calculations further show that there is a pair of topological edge states with opposite chirality and opposite spin polarization at the boundary (Fig. 2c) (see the Supplemental material for more details). Thus, the type-II QSHI can be realized in the monolayer lattice model. Furthermore, in the case where $U(1)$ symmetry is broken ($\mathbf{m} \nparallel \lambda$), the type-II QSHI phase remains robust over a broad parameter range [11].

After realizing the type-II QSHI in the monolayer, we now stack the monolayer lattice into a bilayer lattice model with interlayer ferromagnetic order and weak interlayer hopping $t_z = |t_1|/10$ (Fig. 2a). Below, we use this bilayer lattice model to investigate the properties of stacked type-II QSHIs. The corresponding Hamiltonian is

$$H = \sum_{l=1}^2 H_l + \sum_{\alpha} \left[t_z C_{1,\alpha,j}^\dagger C_{2,\alpha,j} + \text{h.c.} \right], \quad (2)$$

where $l = 1, 2$ and $\alpha = 1, 2$ denote the layer index and sublattice site index per monolayer, respectively. We define the creation operator $C_{l,\alpha,j}^\dagger = (C_{l,\alpha,j1}^\dagger, C_{l,\alpha,j2}^\dagger)$ for sublattice α of the l -th layer. The bilayer lattice model also lacks spin symmetries $\{C_2^\perp | I\}$ and $\{C_2^\perp | \tau\}$ but preserves spin symmetries $\{C_2^\perp | C_{4z}\}$ and $\{C_2^\perp | M_{xy}\}$. Therefore, the bilayer lattice model is also d -wave altermagnetic.

By performing the Fourier transformation, the Hamiltonian (Eq. (2)), in the absence of SOC, can be rewritten in momentum space as $H = \sum_k \psi^\dagger H_k \psi$. The matrix H_k takes the form

$$H_k = \rho_0 \left[\Gamma_k^+ \tau_0 + \Gamma_k^{12} \tau_x + \Gamma_k^- \tau_z - \tau_z \mathbf{m} \cdot \sigma \right] + t_z \rho_x \tau_0 \quad (3)$$

within the basis $\psi^\dagger = (\psi_1^\dagger, \psi_2^\dagger)$, where $\psi_l^\dagger = (C_{l,1k1}^\dagger, C_{l,1k2}^\dagger, C_{l,2k1}^\dagger, C_{l,2k2}^\dagger)$ for $l = 1, 2$. Here, ρ_i and τ_i are the Pauli matrices acting on the layer and sublattice, respectively. The auxiliary functions are defined as $\Gamma_k^{12} = 4t^d \cos \frac{k_x}{2} \cos \frac{k_y}{2}$ and $\Gamma_k^\pm = (t_1 \pm t_2) \cos k_x + (t_2 \pm t_1) \cos k_y$. The eigenvalues of Eq. (3) are given by

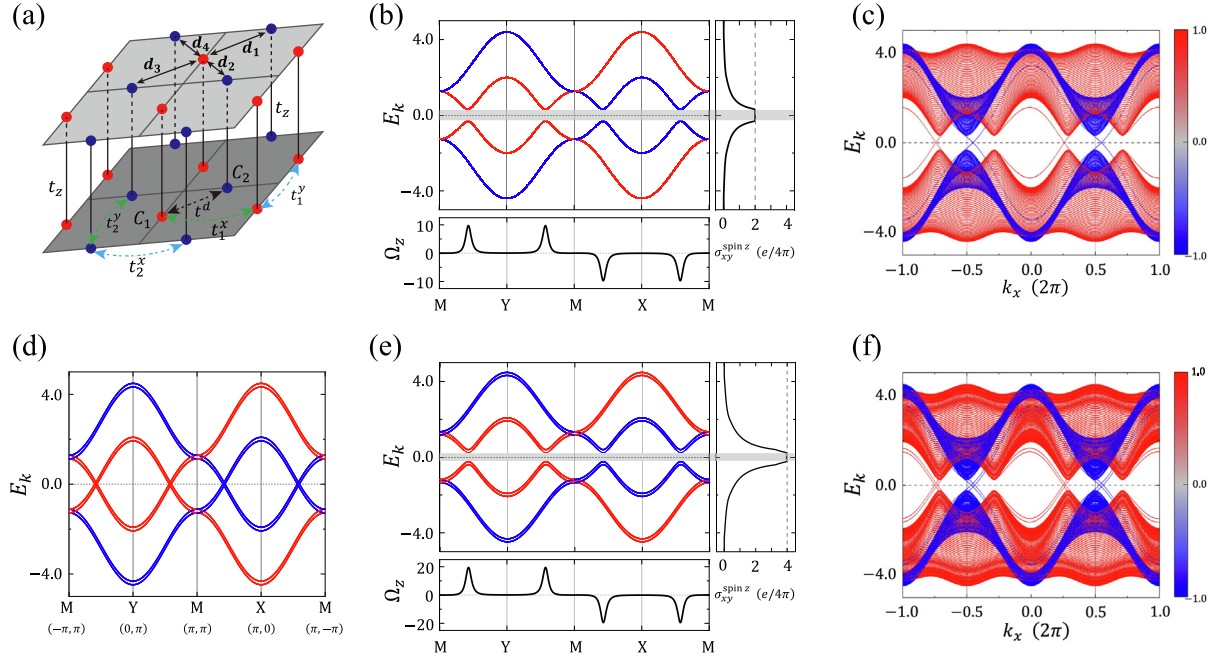


Fig. 2. (a) Stacked bilayer altermagnetic lattice model, where sites with different colors represent sublattices carrying opposite magnetic moments. (b) Band structure, Berry curvature, and SHC of the monolayer altermagnetic model with SOC. (c) The corresponding edge states with s_z projection of the monolayer altermagnetic model, calculated for a 60-unit-cell width altermagnetic nanoribbon with periodic boundary conditions along the x -direction and open boundary conditions along the y -direction. (d, e) Band structures for the bilayer altermagnetic model without ($|\lambda| = 0$) and with SOC. The Berry curvature and SHC of the bilayer altermagnetic model are included in (e). (f) Edge states of bilayer altermagnetic model under SOC. Parameters: $t^d = 1$, $t_1 = -t_2 = 0.8$, $|\lambda| = \lambda_z = 0.1$, $|\mathbf{m}| = m_z = 1.2$ for both monolayer and bilayer altermagnetic models, and $t_z = |t_1|/10$ for the bilayer.

$$E_{r,s,\pm}(\mathbf{k}) = rt_z + \Gamma_k^\pm \pm \sqrt{(\Gamma_k^{12})^2 + (\Gamma_k^- - s|\mathbf{m}|)^2}, \quad (4)$$

where $r = \pm 1$ and $s = \pm 1$ denote the layer and spin indices, respectively.

Here, we set the magnetic moment direction along the out-of-plane ($\mathbf{m} = m_z \hat{z}$). As our symmetry analysis shows, the bilayer lattice model remains d -wave altermagnetic and constitutes a bipolarized spinful Dirac semimetal without SOC, as shown in Fig. 2d. Unlike the monolayer lattice model, there are four pairs of spinful Dirac points at the Fermi level: two pairs with spin-up polarization protected by C_{2x} symmetry, and two pairs with spin-down polarization protected by C_{2y} symmetry. When SOC is considered, $C_{4z}T$ symmetry may drive the bilayer lattice model into a type-II QSHI with a high spin Chern number.

With SOC, the physical symmetry is described by a magnetic group, as the lattice and spin rotations are coupled. Since the bilayer lattice model breaks both C_{2x} and C_{2y} symmetries, all spinful Dirac points open a gap, driving the model from a semimetal to an insulator. As in the monolayer lattice model, we first consider the bilayer lattice model with $U(1)$ symmetry. This bilayer model indeed becomes an insulator (Fig. 2e). Calculations of the Berry curvature show that the two pairs of massive spinful Dirac points with spin-up polarization contribute identical Berry phases (Fig. 2e), so the occupied spin-up states yield a Chern number of 2. Because of the $C_{4z}T$ symmetry, the occupied spin-down states contribute a Chern number of -2 . Consequently, the bilayer lattice model is a type-II QSHI with a spin Chern number 2; the corresponding SHC is $\sigma_{xy}^z = 2e/(2\pi)$, which is confirmed by our calculations (Fig. 2e). Moreover, the type-II QSHI with spin Chern number 2 supports two pairs of topological edge states with opposite chirality and polarization at the boundary (Fig. 2f).

In real materials, SOC inevitably breaks $U(1)$ symmetry. We therefore study whether a type-II QSHI with spin Chern number 2 can still be realized once $U(1)$ symmetry is broken. Our calculations show that the bilayer lattice model without $U(1)$ symmetry

retains its d -wave altermagnetic character (Fig. 3a), and the occupied states of opposite polarization continue to yield opposite Chern numbers of 2 and -2 . Moreover, the SHC remains quantized to $\sigma_{xy}^z = 2e/(2\pi)$, and open-boundary-condition calculations still show two pairs of topological edge states with opposite chirality and polarization at the boundary (Fig. 3b). Even under strong SOC, the type-II QSHI can still be realized (detailed calculation results are provided in the Supplemental material). These results demonstrate that a type-II QSHI with a high spin Chern number can be realized in materials.

3. Candidate material

Guided by the model results, realizing type-II QSHI with a high spin Chern number by stacking requires that the monolayer is a type-II QSHI. In our earlier work, monolayer Nb_2SeTeO was predicted to be such a system. The crystal structure of monolayer Nb_2SeTeO has the $P4mm$ space group, and the corresponding point group is C_{4v} . As shown in Fig. 4a (one layer), it adopts a sandwich structure with Nb-O layer sandwiched between Se and Te sheets, and its magnetic ground state is d -wave altermagnetic [11]. Following the model, bilayer Nb_2SeTeO stacked with intralayer antiferromagnetism and interlayer ferromagnetism retains d -wave altermagnetism. Indeed, our calculations indicate that the bilayer Nb_2SeTeO is a type-II QSHI with a spin Chern number 2 (see the Supplemental material for more details). It is however noticed that there is a tiny overlap between the conduction and valence bands.

To remove this overlap, we apply 0.238% tensile strain along the a -axis and 0.238% compressive strain along the b -axis. Under this small strain, the symmetry lowers to $Pmm2$ (point group C_{2v}). As a result, the Nb atoms with opposite moments are no longer connected by any symmetry, but the net total moment remains zero. Consequently, the strained bilayer Nb_2SeTeO transitions from altermagnetism to unconventional compensated magnetism [15–17]. We then calculated the electronic structure of the strained

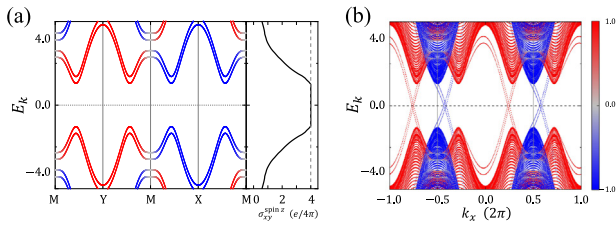


Fig. 3. (a) Band structure and SHC for the bilayer AM model with SOC, when the $U(1)$ symmetry is broken. We set $|\lambda| = 0.2$, $|\lambda_x| = 0$, $|\lambda_y| = |\lambda| \sin \theta$, $|\lambda_z| = |\lambda| \cos \theta$ with $\theta = \pi/10$, namely, $\mathbf{m} \parallel \lambda$, thus the $U(1)$ symmetry is lifted. (b) The corresponding edge states with s_z projection, which are two pairs of topological edge states with opposite chirality and polarization at the boundary. Other parameters are the same in Fig. 2.

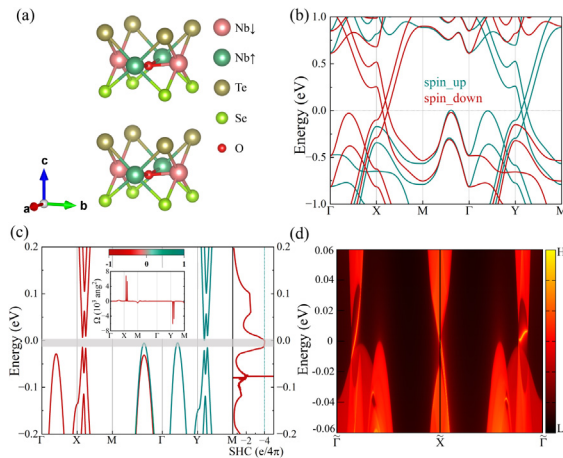


Fig. 4. (a) Crystal and magnetic structures of the bilayer Nb_2SeTeO . (b) Band structure without SOC. (c) Spin resolved band structure and SHC of bilayer Nb_2SeTeO . (d) Edge states for bilayer Nb_2SeTeO cut along the $[100]$ direction.

bilayer Nb_2SeTeO without SOC, shown in Fig. 4b. The strained bilayer is a bipolarized spinful Dirac semimetal without SOC: spin-up and spin-down spinful Dirac points are protected by M_y and M_x mirror symmetries, respectively. Including SOC, the two pairs of spinful Dirac points with spin-up or spin-down polarization contribute Chern numbers of $+2$ or -2 , rendering the strained bilayer Nb_2SeTeO to be a type-II QSHI with a spin Chern number of 2, which is also confirmed by our SHC and edge-state calculations (Fig. 4c and d). Hence, stacking-induced type-II QSHIs with spin Chern number 2 can be realized in both altermagnetic and unconventional compensated magnetic materials. Extending this stacking strategy to multilayers naturally allows type-II QSHIs with a larger spin Chern number.

4. Conclusion

In summary, based on theoretical analysis, we propose that stacking two monolayer type-II QSHIs can give rise to a nontrivial topological phase with a high spin Chern number, in stark contrast to type-I systems. This is demonstrated using a lattice model. Moreover, the magnitude of the quantized SHC and the number of topological chiral boundary states in the bilayer system are both twice those of the monolayer. Symmetry analysis combined with first-principles calculations further predicts that bilayer Nb_2SeTeO hosts such a QSHI phase with a high spin Chern number, and this holds for both altermagnetic and unconventional compensated magnetism. Extending this strategy to multilayer stacks naturally leads to a QSHI with an even larger spin Chern number. Our work

not only deepens the understanding of type-II QSHIs but also offers a route to realizing quantum spin Hall effects with high spin Chern number.

Conflict of interest

The authors declare that they have no conflict of interest.

Acknowledgments

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Author contributions

Peng-Jie Guo, Fengjie Ma, and Zhong-Yi Lu conceived the study and proposed the idea. Chao-Yang Tan and Panjun Feng contributed equally to this work. Chao-Yang Tan performed the model study and Panjun Feng performed the first-principles calculations; all the authors discussed the results and commented on the manuscript.

Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.scib.2026.04.018>.

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